

Time : 3 Hours]

[Total Marks : 60

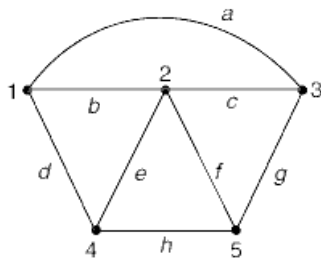
Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

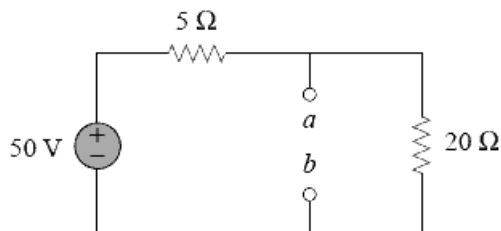
Q1. Objective type questions. (Compulsory Question)

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- 1 Identify which of the following is NOT a tree of the graph shown in Fig

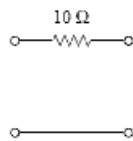


- (a) begh (b) defg (c) adfg (d) aegh
- 2 The number of node equations required in nodal analysis is equal to:
A) Total number of nodes B) Total branches
C) Total nodes minus one D) Total loops
- 3 A network has 12 branches and 8 independent loops. How many nodes are there in the network?
(a) 19 (b) 17 (c) 5 (d) 4
- 4 The Norton current at terminals a and b of the circuit in Fig. is:



- (a) 10 A (b) 2.5 A (c) 2 A (d) 0 A
- 5 A load is connected to a network. At the terminals to which the load is connected, $R_{Th} = 10 \text{ ohm}$ and $V_{Th} = 40 \text{ V}$. The maximum power supplied to the load is:
(a) 160 W (b) 80 W (c) 40 W (d) 1 W
- 6 The time constant of an RC circuit is given by:
A) R/L B) L/R C) RC D) $1/RC$

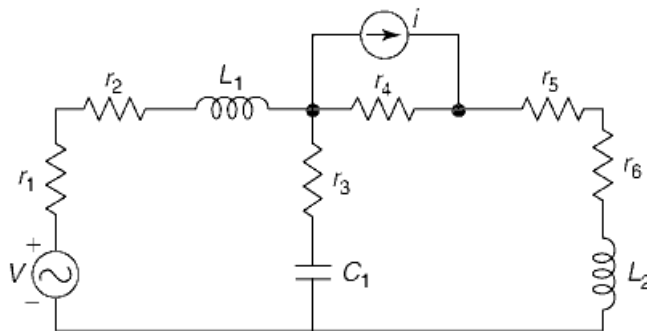
- 7 The solution of a **non-homogeneous linear differential equation** consists of:
 A) Only complementary function B) Only particular integral
 C) Complementary function + particular integral
 D) Initial conditions only
- 8 In a series RLC high Q circuit, the current peaks at a frequency
 (a) equal to the resonant frequency (b) greater than the resonant frequency
 (c) less than the resonant frequency (d) none of the above.
- 9 For a series resonant circuit at low frequency, circuit impedance is _____ and at high frequency circuit impedance is _____.
 (a) capacitive, inductive (b) inductive, capacitive
 (c) resistive, inductive (d) capacitive, resistive
- 10 For the single-element two-port network in Fig , B of transmission parameter is:
 (a) 0 (b) 5 (c) 10 (d) 20



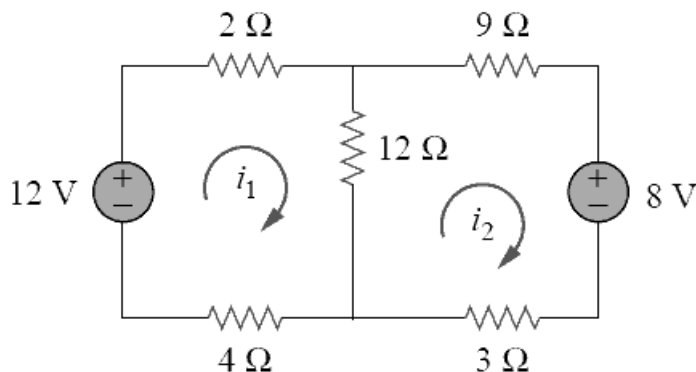
- 11 An RC circuit has $R = 2 \text{ ohm}$ and $C = 4 \text{ F}$. The time constant is:
 (a) 0.5 s (b) 2 s (c) 4 s (d) 8 s
- 12 The difference between the half-power frequencies is called the:
 (a) quality factor (b) resonant frequency (c) bandwidth (d) cutoff frequency

Q2. Solve the following.

- A) For the circuit shown in Fig., draw the oriented graph and write (a) incidence matrix, (b) tieset matrix, and (c) cutset matrix. 6



- B) Calculate the mesh currents i_1 and i_2 in the circuit of Fig. 6

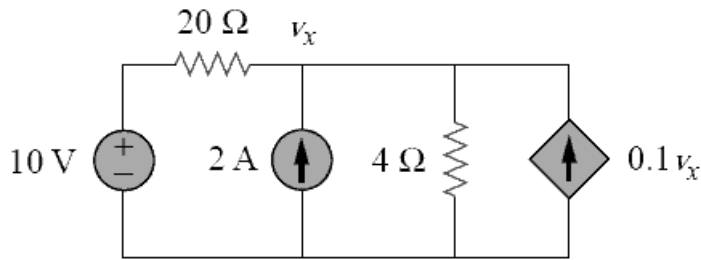


Q3. Solve the following.

- A) Explain with its steps Maximum power transfer and Tellengen's Theorem 6

B) Use superposition to find V_x in the circuit in Fig.

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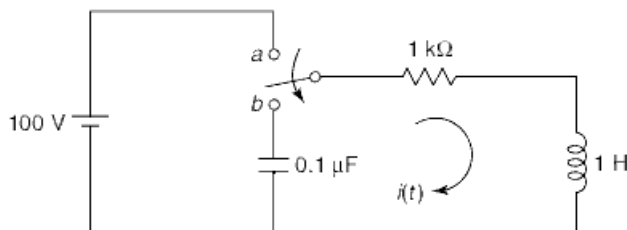


Q4. Solve Any Two of the following.

A) In the network of Fig, the switch is changed from the position 'a' to 'b' at $t = 0$. Solve for:

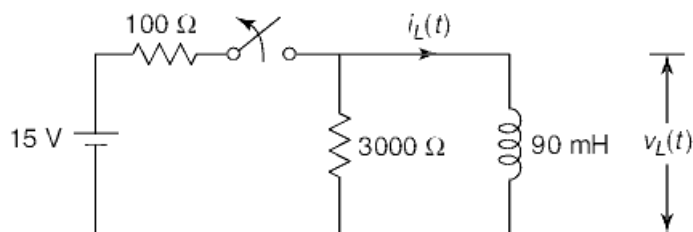
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i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t = 0^+$.



B) For the network shown in Fig. 10.92, steady state is reached with the switch closed. The switch is opened at $t = 0$. Obtain expressions for $i_L(t)$ and $V_L(t)$.

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C) Obtain Voltage across capacitor and resistor for Series Resistor capacitor circuit.

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(Note: The switch is closed at time $t = 0$. The capacitor is initially uncharged.)

Q5. Solve Any Two of the following.

A) Explain the concept of a balanced three-phase system. Derive the relations between line and phase voltages and currents for a star-connected load. Also draw its phasor diagram.

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B) Explain the resonance phenomenon in a series RLC circuit. Derive the expression for resonant frequency, bandwidth, and quality factor (Q).

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C) A parallel resonant circuit has $R = 100\text{ k}\Omega$, $L = 20\text{ mH}$, and $C = 5\text{ nF}$. Calculate ω_0 , ω_1 , ω_2 , Q, and B.

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Q6. Solve Any Two of the following.

A) Derive the Z-parameter equations of a two-port network.

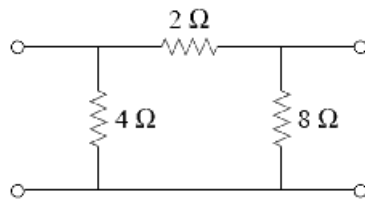
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B) With neat Diagram Explain active bandpass filter. Derive its transfer function and draw its ideal and practical frequency response.

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C) Obtain the y parameters for the network shown in Fig.

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***** End *****